



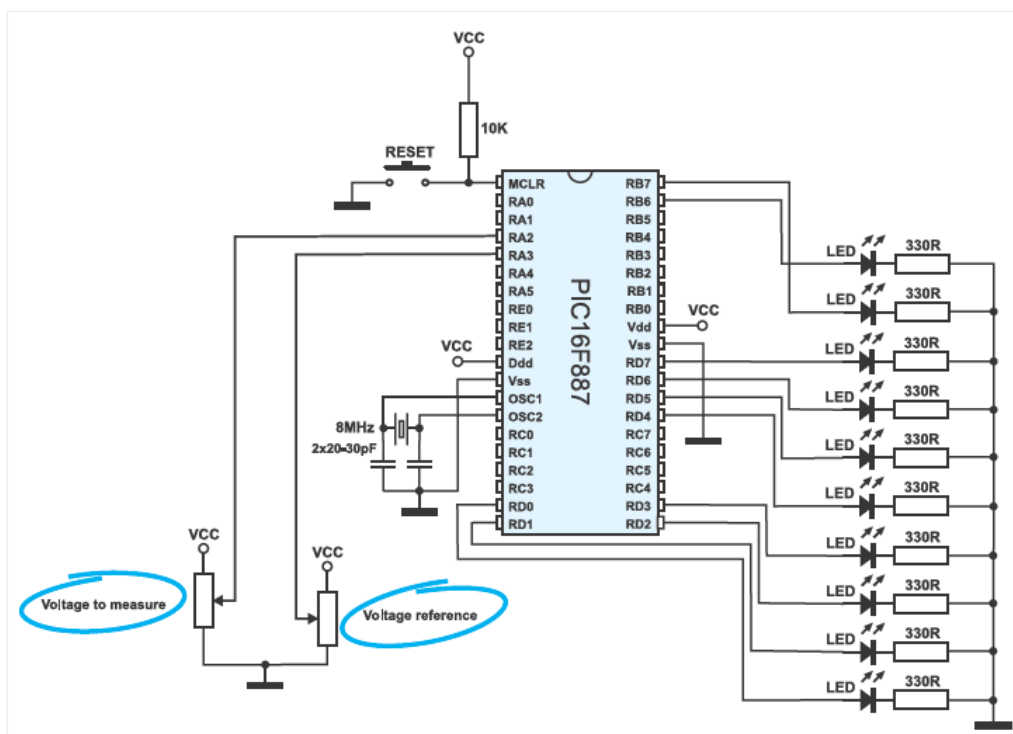
4.9 EXAMPLE 7

MIKROELEKTRONIKA

Using A/D converter

The PIC16F887 A/D converter is used in this example. Is it necessary to mention that everything is rather simple?! A variable analog signal is applied to the AN2 pin, while the 10-bit result of conversion is shown on ports B and D (8 LSBs on port D and 2 MSBs on port B). GND is used as negative voltage reference Vref-, while positive voltage reference is applied to the AN3 pin. It enables voltage measurement scale to 'stretch and shrink'.

In other words, the A/D converter always generates a 10-bit binary result, which means that it detects a total of 1024 voltage levels ($2^{10}=1024$). The difference between two voltage levels is not always the same. The less the difference between Vref+ and Vref-, the less the difference between two of 1024 levels. As seen, the A/D converter is able to detect slight changes in voltage.



```

1  /*Header*****//
2
3  unsigned int temp_res;
4
5  void main() {
6      ANSEL = 0x0C; // Pins AN2 and AN3 are configured as analog
7      TRISA = 0xFF; // All port A pins are configured as inputs
8      ANSELH = 0; // Rest of pins is configured as digital
9      TRISB = 0x3F; // Port B pins RB7 and RB6 are configured as
10     // outputs
11     TRISD = 0; // All port D pins are configured as outputs
12     ADCON1.F4 = 1; // Voltage reference is brought to the RA3 pin.
13
14     do {
15         temp_res = ADC_Read(2); // Result of A/D conversion is copied to temp_res

```

```
16 PORTD = temp_res; // 8 LSBs are moved to port D
17 PORTB = temp_res >> 2; // 2 MSBs are moved to bits RB6 and RB7
18 } while(1); // Endless loop
19 }
```

In order to make this example work properly, it is necessary to tick off the ADC library in the **Library Manager** prior to compiling:

- ADC



4.9 Example 7 by MikroElektronika is licensed under a Creative Commons Attribution 4.0 International License, except where otherwise noted.